



Thep Watana Co., Ltd.

382 Bangpoo Industrial Estate Sukhumvit Rd., Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66-2721-3510

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SAFETY DATA SHEET

Company Identification

Thep Watana Co., Ltd. Address: 382 Bangpoo Industrial Estate.

Sukhumvit Road, Preaksa, Muang District, Samutprakarn 10280, THAILAND

Phone: +66(0)-2721-3510;

Fax: +66(0)-2721-3515

1. CHEMICAL IDENTIFICATION

Product Identification

Product name: fipronil 5% W/V SC

Product use: Insecticide

Relevant identified uses of the substance/mixture and uses advised against:

Used to control damping-off, root rot and stem rot caused by *Pythium* and *Phytophthora*. It is used on golf course turf, cotton and nursery ornamentals, non-bearing citrus, non-bearing coffee, and mixed with potting soil for greenhouse use. It is also used as a seed treatment on barley, beans, corn, cotton, peanuts, peas, sorghum, soybeans, safflower and wheat. Seven states hold special local need registrations for use in tobacco transplant floatbeds.

2. Hazard Identification

Acute Health Effect

Swallowed:

Harmful if swallowed.

Eye:

Will irritate eyes. May cause weeping and redness.

Skin:

Will irritate skin. Fipronil does not readily penetrate the skin.

Inhaled:

Harmful if inhaled.

Other

Symptoms of overexposure include respiratory difficulty, decreased reflexes, lethargy, tremors, and in severe cases, convulsions. Repeated exposure at dose levels that are not acutely toxic may result in symptoms of toxicity.



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3. COMPOSITION/INFORMATION ON INGREDIENT

Chemical name: ($\pm$)-5-amino-1-(2, 6-dichloro- α , α , α -trifluoro-*p*-tolyl)-4-trifluoromethylsulfinylpyrazole-3-carbonitrile (IUPAC)

Common name: fipronil

CAS Number: 120068-37-3

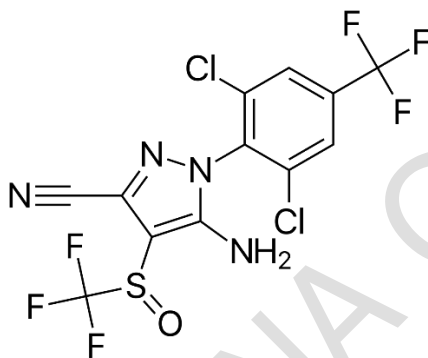
Ingredients and composition: Active ingredient – Fipronil 50 g/L

Other ingredient to 1 L

Chemical family: Phenylpyrazole

Formula: $C_{12}H_4Cl_2F_6N_4OS$

Molecular weight: 437.141 g/mol



Structural formula:

4. FIRST AID MEASURES

Description of first aid measures

In case of swallowed:

If swallowed, immediately contact a doctor or Poisons Information Centre, and follow the advice given. Keep under medical supervision.

In case of eyes contact:

Immediately irrigate with copious quantities of water for at least 15 minutes. Seek immediate medical assistance.

In case of skin contact:

Remove contaminated clothing. Wash affected areas with plenty of soap and water.

In case of Inhaled contact:

Remove victim to fresh air. Seek medical assistance.

First Aid Facilities:

Ensure washing facilities are available, including eyewash facilities.

Advice to Doctor:

Fipronil is a reversible gamma-aminobutyric (GABA) receptor inhibitor. During intoxication it will induce neurological stimulation with possible convulsions. Treat symptoms. No specific



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antidote known. Phenobarbital, and to a lesser extent, benzodiazepines, have been shown experimentally to be effective in preventing convulsions induced by fipronil. Due to slow absorption of fipronil through the gut, symptoms of intoxication may be delayed several hours to one day. Absorption may be decreased by the use of gastric lavage, saline purgative and activated charcoal (possible entero-hepatic recirculation).

Continue monitoring – due to the slow elimination of the compound.

5. FIRE FIGHTING MEASURES

Fire and Explosion hazards:

Extinguishing Media: Carbon dioxide, water spray, dry agent, foam.

Special Fire Fighting Equipment Procedures:

Firefighters should wear full protective gear, including self-contained breathing apparatus (AS/NZS 1715, 1716). If possible and without risk, remove intact containers from exposure to fire. Otherwise, spray unopened containers with water to keep cool. Whenever possible, contain fire-fighting water by bunding area with sand or earth to prevent it entering any bodies of water.

6. ACCIDENTAL RELEASE MEASURES (SPILLAGE)

Procedure for spillage and leakage:

Avoid contact with the spilled material or contaminated surfaces. When dealing with spills do not eat, drink or smoke and wear personal protective clothing and equipment as described in PERSONAL PROTECTION section. Prevent spilled material from entering drains or watercourses. Contain spill and absorb with earth, sand, clay, or other absorbent material. Collect and store in properly labelled, sealed drums for safe disposal. Deal with all spillages immediately. If contamination of drains, streams, watercourses, etc. is unavoidable, warn the local water authority.

Neutralizing Chemicals:

Hydrated lime.

7. HANDLING AND STORAGE

Good industrial practice in housekeeping and personal hygiene should be followed.

Handling & Storage Precautions:

Keep out of reach of children. Store in the closed, original container in a dry, cool, well-ventilated area, out of direct sunlight. Store in a locked room or place away from children, animals, food, feedstuffs, seed and fertilizers.



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8. EXPOSURE CONTROL/PERSONAL PROTECTION

Engineering Controls:

Control process conditions to avoid contact. Use local exhaust ventilation during manufacture.

Use in a well-ventilated area only.

Personal Protective Equipment

Eyes Protection:

Safety goggles

Skin Protection:

Cotton overalls buttoned to the neck and wrist (or equivalent clothing)

Hand Protection:

Elbow-length PVC gloves

Respiratory Protection:

Do not breathe vapour or spray. Wear an organic cartridge respirator if inhalation is possible.

Other:

Wash hands after use. After each day's use, wash gloves, goggles, respirator facepiece and contaminated clothing.

9. PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Physical stage:	Liquid
Color:	White suspension
Odor:	Vegetable oil smell
pH:	4.0 – 7.0
Boiling point:	Not determined
Flash point:	Non-flammable
Flammability Limits:	Not applicable
Ignition temperature:	410 °C
Vapor pressure:	3.7×10^{-4} mPa at 25 °C
Solubility in water:	Disperses in water
Specific gravity:	1.05 – 1.07 at 20 °C (H ₂ O=1)

10. STABILITY AND REACTIVITY

Reactivity Hazards: Stable under normal conditions of use. No dangerous reaction known under normal conditions of use. Avoid extreme heat and fire. Toxic thermal decomposition products may include oxides of carbon, sulphur and nitrogen, and compounds of chlorine and fluorine.



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11. TOXICOLOGICAL INFORMATION

Information on Acute toxicity for end product, fipronil:

LD50 (Oral, rat):	1,127 mg/kg
LD50 (Dermal, rabbit):	> 2,493 mg/kg
LC50 (Inhalation, rat, 4h):	> 0.5 mg/L
Skin irritation (rabbit):	Practically non-irritating to skin of rabbits.
Eye irritation (rabbit):	Slightly irritating to eyes of rabbits.
Skin sensitization:	Not a skin sensitizer to guinea pigs.

12. ECOLOGICAL INFORMATION

Toxicity

Adopt good working practices, so that the product is not released into the environment.

Data referred to the technical, fipronil:

Fish – Acute toxicity:

Rainbow trout,

LC50 (96 h): 248 µg/L

Bluegill sunfish,

LC50 (96 h): 85 µg/L

European crap,

LC50 (96 h): 430 µg/L

Daphnia magna,

LC50 (48 h): 0.19 mg/L

Birds – Acute toxicity:

Bobwhite quail,

LD50: 11.3 mg/kg

LC50 (5 days): 49 mg/kg diet

Mallard ducks,

LD50: > 2,000 mg/kg

LC50 (5 days): > 5,000 mg/kg diet

Pheasants,

LD50: 31 mg/kg

Red-legged partridges,

LD50: 34 mg/kg

House sparrows,

LD50: 1,120 mg/kg



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Pigeons,

LD50: > 2,000 mg/kg

Algae – Acute toxicity:

Selenastrum Capricornutum,

EC50 (120 h): > 0.16 mg/L

Bee – Acute toxicity:

Highly toxic to honeybees both by direct contact and by ingestion. However no risk to bees when used as a soil or seed treatment.

13. DISPOSAL CONSIDERATIONS

Waste Disposal Methods:

Triple or preferably pressure rinse containers before disposal. Add rinsings to spray tank. Do not dispose of undiluted chemicals on site. Break, crush, or puncture and bury empty containers in a local authority landfill. If no landfill is available, bury the containers below 500 mm in a disposal pit specifically marked and set up for the purpose, clear of waterways, desirable vegetation and tree roots. Empty containers and product should not be burnt.

14. TRANSPORT INFORMATION

The product is classified as a Dangerous Good under the Australian Code for the Transport of Dangerous Goods by Road and Rail: PESTICIDE, LIQUID, TOXIC, N.O.S. (contains fipronil), Class: 6.1, PG III, UN 2902. Must be stored and transported according to local, State/Territory and Federal regulations.

15. REGULATORY INFORMATION

EPCRA SARA Title III Classification

Section 311/312 Hazard Classes: Acute Health Hazard
Chronic Health Hazard

Section 313 Toxic Chemicals: None

Unbreakable packaging; put breakable packaging into closed unbreakable container. Do not transport with food and feedstuffs.



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16. OTHER INFORMATION

This information is provided in good faith but without express or implied warranty. Buyer assumes all responsibility for safety and use not in accordance with label instructions.

THEP WATANA CO., LTD.